Impromptu Google Meet Meeting - December 22

VIEW RECORDING - 59 mins (No highlights)

@0:01 - **Tillman Sherwood**

Yes, I do. Perfect. Hello. Nice to meet you. Good. Yeah. I'm just going to go to Christmas markets after this.

So looking forward to enjoying that. Oh, nice. Which one? In Mitte, the one at Kufustendam.

@0:21 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Oh, Kudam.

@0:23 - **Tillman Sherwood**

Yeah, yeah, exactly. Yeah. Cool, cool. Nice. How about you? How are you doing? I'm doing great.

@0:30 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Also, preparing for Christmas. So what would you like to do? Today is kind of like a free day. Not a lot of people available to work.

@0:41 - **Tillman Sherwood**

Yeah.

@0:42 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Thank you so much for taking the time. I want to ask, is it okay if I activate my note taker to take notes?

Of course.

@0:51 - **Tillman Sherwood**

Yeah, go for it.

@0:52 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Thanks. Just, um, yeah, I'll, I'll get like a quick intro. Introduction about myself. So my name is Mouaz. I moved to Berlin 12 years ago, studied computer science in the TU Berlin.

I've been working in the startup ecosystem for the last 12 years, mainly building SaaS, worked in different domains, but I've been into the topic of sustainability in the last couple of months.

And I joined Carbon13. It's a startup accelerator in Berlin and Cambridge. And I met my two current colleagues right now, Olaf and Rene.

And we're working on the topic of sustainability in discrete manufacturing. Yep. And we're interviewing people that are working in that domain to learn more about this topic.

And we're planning to ... ... Generate like a white paper by February and I would be happy also to share some of the results with you.

Yeah, sure. Just want to make sure like how much, how much time we have, like would you, would you have for this?

@2:18 - **Tillman Sherwood**

I can have, I, yeah, really like 40 minutes, maybe. I mean, I don't know how, how relevant the experience that I had, but let's see.

I'm pretty flexible. So if you have questions.

@2:33 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I just want to make sure. So we're basically, we're just trying to understand more about this topic from people that are active in it, like especially mechanical, mechanical and electrical engineers, people who work with CAD, works in discrete manufacturing.

And we want to see. Where sustainability sits in the design phase. I don't know, you probably know this, but 80% of products impacts lies in the design phase usually.

@3:12 - **Tillman Sherwood**

Okay. Yeah, interesting, because maybe a bit of an introduction to GaiaStar. I am no longer with the company, I left in September.

But they are making these 3D printed clay cups that you can see here. And what's special about them is that they're very thin walled.

So it's like really kind of supposed to mimic a paper cup. But it's made out of clay. So it's fully just dirt and water and some salt.

And it's glazed also, so it's kind It's of like glass as well. It's like, it's a ceramic, basically. It's a technical ceramic, very thin walled.

It's very lightweight and it provides like a unique luxury of drinking experience. So the main customers were like hotel brands and high end cafes in cities.

Yeah. And then at the end, you can, like, they were lasered with the brand name of, or custom logos, custom, yeah, labels.

So this one was V Cafe. So it's a custom label from a cafe in Switzerland that we worked with.

Um, yeah. And so this is all made from clay. So the manufacturing process was quite challenging, I would say.

Um, and because it was a unique manufacturing process, um, from the design also, I mean, maybe let's start with the design.

Cause you just said that the design is a very important phase. Um, it was challenging because they need to be like, ideally they need.

They be stackable, they need to be robust, like strong, but they are pretty strong for, because ceramics are quite a strong material in general, but they also needed to have no sharp corners or anything that could build stress and tension and crack then upon drying.

So, and also that's the most important thing. So these are made with molds. So the molds are made of plaster and basically the clay is like injected into the spinning mold and it climbs up the

wall and then the plaster mold absorbs the water and then out comes the dry clay and then the clay needs to be fired.

So that's the problem, I think, is that the mold needs to be very, like have these smooth corners and I think that was like the biggest challenge for them.

And also to have like many. Manufacturing precision that we were able to create these very thin walled structures because there needs to be a positive and a negative part that are moving always so like there was something that came in and injected the clay and while it was moving like this came out so yeah so it was just a challenge from that perspective but I think at the at the design stage there was it's really like solid clay there's not that much flexibility I mean there's different products they had like smaller products like this or like ice cream cups um but really it's just a thin walled clay cup um and then from the manufacturing there was actually a lot that could be done to to make it more sustainable but I'll pause there if there's any questions or things that you'd like me to dive into that's uh that's really interesting and uh yeah I see like how challenging I think uh with this this week less

@7:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Like under discrete manufacturing? I think so.

@7:03 - **Tillman Sherwood**

Yeah, because we, you know, we had to make every, um, or let me refresh my memory on. Um, yeah, I don't know.

I think you'd have to check me on that because I guess we were trying to go into mass, mass product manufacturing.

I guess it would be more serial because I think discrete manufacturing is specific for like assembly of products and like creating individual products with individual components.

So, um, yeah, I mean, these were, it's more, I think, closer to serial manufacturing because every, every product is kind of unique, but it's really more mass manufacturing.

Um, yeah,

@8:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I was just wondering, but yeah, I think there's only one component, which is the clay, and then you do more processing on it.

@8:08 - **Tillman Sherwood**

We were trying to get to mass manufacturing, and I think we were kind of there, but it's not trivial.

It's pretty difficult to do. You need like a high demand. Yeah, I mean, we had the demand. It's just the amount of process steps, actually.

So there was 10 different process steps. And each process step took a non-trivial amount of time. I can actually share my screen and share some of the data that I collected.

part of my role was, I mean, I was really broadly as an engineer applied, so I don't like, I can speak to a lot of things, I think, in the company, but part of my role was like collecting data and visualizing it.

So first we printed the cups, I kind of already described that process, then we fired them. Then we checked that they were okay after firing and removed them from the kilns, then we glazed them, then we glazed-fired them, then we did another QC step, then we branded them with the logos as a laser branding, then we packaged them, and then we shipped them.

Oh, cool.

@9:25 - **Mouaz Al Qudsi (mouazq@gmail.com)**

And those are the number of cups that's been... Yeah, yeah.

@9:30 - **Tillman Sherwood**

And you see, like, during when I started, there was a very small amount, and we ramped it up pretty quickly during the first six months.

But, yeah, I mean...

@9:49 - **Mouaz Al Qudsi (mouazq@gmail.com)**

It's, like, nice.

@9:50 - **Tillman Sherwood**

So, yeah, mean, printing... In order for printing to happen... We needed there to be clay mixed also before, and clay needed to be prepared and put into chambers, little cartridges for the printer.

So that was also a process step. And I think these combined combination of all these different process steps was really what was the challenge to increase production.

Nice.

@10:35 - **Mouaz Al Qudsi (mouazq@gmail.com)**

And this product was made for more to say optimized for sustainability. I think that's the main goal. this was one of the things.

@10:44 - **Tillman Sherwood**

Yeah. first of all, it's plastic free. And it's reusable. It was sold as a single use product. product. I

But yeah, it's technically, so as a single-use product, technically, you could throw it away, and it's a ceramic, so it has zero waste, and you put it in landfill, or we were even coming up with ideas to grind it up and reuse it in our process, or in, for example, like making pavements look prettier, or here I can show you maybe the website.

But yeah, so that was one aspect, so it was plastic-free, single-use, but also it's so strong that you can use it multiple times.

There's even dishwashers save someone, and we were trying to get certification for that, and then another part was that it was supposed to be with local material sourcing, and yeah, otherwise we were trying to create it so that going forward, we had a lower a

More energy output, a higher yield, and then we were also tracking the life cycle analysis, basically, to create.

@12:10 - **Mouaz Al Qudsi (mouazq@gmail.com)**

What kind of tools you were using for LCA?

@12:16 - **Tillman Sherwood**

We were trying to do kit certification, for example, like dishwasher safe. So we were sending it to certification bodies.

I'm not sure exactly which ones. I believe there was one in Germany and a couple in the US that we were trying to work with to get these.

But we were also using a product life cycle manager called Arena. But we were using this for tracking different bills of materials and things and documentation mostly.

@12:50 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Nice. So those kinds of things, I think you as a designer would come after for a separate different department.

I think the LCA management. And then. Thank

@13:01 - **Tillman Sherwood**

Yeah, it wasn't necessarily tracking the life cycle of the cups.

@13:14 - **Mouaz Al Qudsi (mouazq@gmail.com)**

My question is, I think you've been working for how long you've been working with product design and engineering, and is this the main thing you have worked on?

@13:25 - **Tillman Sherwood**

Yeah, I mean, I also was part of different uni groups, and I was part of a small team where I worked as a product design mainly, building a camera concept for an infrared high resolution camera for reading paintings and scientific measurements.

And now I'm actually going into my next role is in medicine, in a production company that is making medical products.

let's see. I'm going to be working on the product development team, actually, so there'll be even more CAD than in my previous role.

So, yeah, I have some experience also in uni in different groups, like Formula Student, where you make a race car as part of a challenge with a big team of engineers.

Yeah, so I have a few different experiences to pull from, but my main working experience has been at GaiaStar.

@14:28 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Has been what, sorry? It has been at this company that makes the cups. Nice.

@14:36 - **Tillman Sherwood**

Oh, sounds exciting.

@14:38 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Also like with the camera concept design for Fred and Reading. So was it like a prototype there when you worked on that?

Yeah, yeah, it was.

@14:46 - **Tillman Sherwood**

So the idea that they were a hyperspectral imaging company. They work in Scotland, or they're based in Scotland. It's a very small team.

But they have a very niche industry. Thank you. industry where they basically use hyperspectral images, where you can see the complete wavelength, continuous, across the spectrum, not just the visible wavelength, like normal images, but you can also see in infrared, and you can use this for creating special, yeah, you can, like, you can, like, learn more about something like, like, you'd see, like, a fruit where it might be rotten in a couple days, or you can see when, if meat is off, using the infrared wavelength, so my application was to use it for imaging paintings, so you could see below the surface layer of paintings, and often old painters like Picasso, and, and so on, they used, they made multiple paintings on one canvas, because they didn't have the money, and so using infrared technology, you can see below the paintings, and you can see what their original intentions were, but that was, so that was the application, but the design was,

Maybe I can show some pictures as well. It was to create a design that incorporated the sensor, the infrared sensor, into a scanning system to make it high-resolution.

So here, just a minute. Here, I'll share my screen.

@16:40 - **Mouaz Al Qudsi (mouazq@gmail.com)**

This is kind of my portfolio.

@16:47 - **Tillman Sherwood**

So I worked with the manufacturer also, too. Oh, wait, sorry, sorry. Yeah, to come up with concepts that were actually manufacturable, and yeah, I created engineering drawings and different versions, but basically it was the optical system, the electrical system, and mechanical system.

So that was like more of the modeling phase?

@17:21 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Yeah, yeah.

@17:22 - **Tillman Sherwood**

And also prototyping. I did some prototyping as well. Oh, okay.

@17:27 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So you built like a small prototype, and then you will send it to a manufacturer, and you will give them the 3D design and the bomb, I assume?

@17:37 - **Tillman Sherwood**

Yeah, yeah, exactly. So I give them these engineering drawings, and we would go there and ask them how manufacturable it is, but when I say prototyping, I built like a box with this lens in it to see if the image shows up clearly, at what distance.

And then I also worked with The electronics, like we made this PCP through a Chinese supplier, and I also ordered this, you can hardly see it, but in the back there's this PLC, it's a programmable logic controller to control like the actuators and sensors and things like that.

So I worked with these electrical tools to try to create these things and also test if it works with these motors and sensors and select components that way.

@18:33 - **Mouaz Al Qudsi (mouazq@gmail.com)**

And you did the testing also on some CAE, CAM tools? No, it was mostly like hands-on, was redboard and white.

@18:43 - **Tillman Sherwood**

Oh no simulation, it wasn't old? No.

@18:46 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I think it's hard to simulate this kind of products.

@18:49 - **Tillman Sherwood**

Yeah.

@18:51 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Nice, nice. Cool. Seems like I have a really great experience with like... Yeah, love using CAD.

@19:04 - **Tillman Sherwood**

It's fun.

@19:07 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I mean, you worked mainly for GaiaStar, but you worked on this infrared camera before. So what do you think sustainability sits in all those organizations you have worked for?

Is there anyone more responsible for it? Or was it a topic that doesn't sit in the organization usually?

@19:36 - **Tillman Sherwood**

I mean, here it was honestly less, I hesitate to use the word relevant, but I don't think they need these machines to be made.

And like, they want to make it for a cheap price. And so I think there wasn't really a lot of consideration done for sustainability.

you. Thank Thank Thank With this project specifically, but with GaiaStar, here's for example, I mean with GaiaStar it was hard I think because it was a startup that, yeah I can stop presenting now, there was a startup that was like trying to, or they are a startup that are trying to grow, and so I think it's hard to grow while also keeping an eye on sustainability and where you're sourcing your materials for example, but of course sustainability is at the forefront of their mission also, so I mean their mission is to reduce the use of single-use plastics, and so it's difficult to you know be able to make things, like that's what I'm trying to say I think is that in startups you often are making things very fast, and you're making them with very tight timelines and also you know at a good budget,

But when things are, you know, a little bit cheaper or faster, then they often are, you know, maybe less sustainable because they come from across the country or across the world.

Yes, makes sense. I think an issue with GaiaStar was actually, so we did have a person that was responsible for certification and looking into getting these certifications, working on the, getting the material tested, and working on, like, the product image, basically.

And there was another person in charge of, like, marketing generally. Like, so, yeah, she was the product designer. So she was product designer, but she also worked on, like, the image of the product.

So, like, certification and so on. And there was also a marketing person, specialist, who was working. . with projecting that image to companies.

Nice.

@22:04 - **Mouaz Al Qudsi (mouazq@gmail.com)**

And I assume this is because also the selling point of Gaia Star was also to make a sustainable product.

@22:10 - **Tillman Sherwood**

Exactly. So it was a challenge. I kind of wrote down rough, rough metrics that for me as an engineer that were a sign of sustainability.

I think so we were trying to attract material consumption. We needed a lot of clay. And so this was kind of always a question, like, can we make the cups thinner?

Can we make the cups lighter? Because then if we make them thinner and lighter, then we use less material.

And using less material is better because, you know, it's more efficient. We're also always tracking yield and output, like I showed you, because higher yield means less things that we throw away pretty simply.

And higher output means... means... Also that our system is running efficiently, and then we were also tracking energy output, for example, so these kilns are very, very costly, they're electric, so they take a lot of energy to fire them, and a lot of energy goes into the cups, so we actually did a comparison study with, like, comparing roughly the energy output of one of the cups compared to, like, a plastic water bottle, for example, And we discovered that we would need to move towards continuous kilns, which are, so I can show again, you can see in the image here, that these kilns here are, you know, just open spaces, you can, you know, you can fire them, it's called, like, a hearth kiln, so they're, like, electrical windings,

And you just push a bunch of cups in there, and then you fire it, but it's actually less efficient than if you have a continuous kill, because a continuous kill, you can always be having it loaded, and it doesn't take time to heat up and cool down.

So per item, actually, ironically, mass production is less environmentally costly, less energy costly, because it's always on, and the actual energy that goes into each item is less than if you fire it individually, if that makes sense.

@24:38 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Yeah, and that's really interesting, because that's what I thought about when you first talked about it, that manufacturing those cups takes a lot of energy, so it's like you have a kind of trade-offs between that at the end.

At end, the cab doesn't have a bad environmental impact after it's been out on the market, but to remanufacture it requires a lot of energy, so I think some kind of measurement for, I think if you want to compare it to other cab manufacturing and how are being recycled and their environmental footprint after they're out of the market, that also would be interesting to do with Scope 3.

I think there's this graph where it breaks even with other maybe non-recyclable cabs or other cabs that are recyclable, but also it's really costly to recycle them.

Yeah. And this is a matter of like how much I think you use it at the end. So if you use it for a year, then the carbon footprint of this cup would be better than using other cups.

don't know. Exactly.

@26:15 - **Tillman Sherwood**

Yeah, exactly. That was the idea. But we had no way of measuring this. So it was often challenging to get these metrics to ask customers how many times are you reusing it.

And we were actually working with cafes to get the cafes to take these metrics, like how many customers are bringing their cups back, how many customers are buying them with the paper lid or and so on to get an idea of of the long term, long term scope, because long term, we were trying to make a continuous manufacturing, which would, like I said, would increase yield, would reduce the energy output per cup, also reduce material consumption.

So we were always working to remove process steps and implement more efficient processes.

@27:09 - **Mouaz Al Qudsi (mouazq@gmail.com)**

There's like an interesting line you mentioned to me. you mentioned that can we make the cup thinner and the cup lighter?

And it seems to me this involves some kind of a need for simulation to see the durability, the robustness of the cup.

How did you go about this topic, about those little attributes that you have to optimize? What did you use?

@27:41 - **Tillman Sherwood**

Honestly, the old fashioned way, we did a trial and error. So we tried to print it as thin as we could and then we stress tested it, used it a bit.

We actually had someone that was physically using a strain gauge. to pull apart the cups and it would show how much strength it took to destroy the cup and then we use that to compare you know how at different thicknesses how the strength goes down but yeah simulation I think would be a challenge because in ceramics I don't know how how easy it is to simulate that or like how

accurate the simulation models are in current FEM I mean but also I think there's different like factors like the moisture content or the way it was dried the way it was fired if it's like on the corner of the kiln maybe the temperature profile was different than in the center of the kiln so there's also different challenges that we had that could maybe you know make one cup weaker than the other cup but they actually were made in for us the same way for example like we had a problem that

That some of the cups were actually turning out oval. Um, this one's a little bit oval, but it's hard to see probably because of the way that they were cooled.

Um, and so this was also a challenge when you go too thin. And so I think to simulate, this would have been like a very considerable task.

And I don't know if, uh, we had the time for that. So we, we just did it physically actually.

Yeah.

@29:27 - **Mouaz Al Qudsi (mouazq@gmail.com)**

It's interesting because yeah, so physically was less time consuming than doing it, uh, on the, on the computer, the opposite.

Like the idea of simulation is that you don't want to go through a lot of iterations, physical iterations, because they're time consuming.

So you do it on the computer, but this is the opposite. Um, but this is only because this is ceramic and it's a special case.

And, uh, there's a lot of conditions involved in manufacturing it. Would you say this, uh, for other. It's really hard to kind of also conduct those simulation tests or, you know, yeah, with the way the CAD systems, or sorry, what was the last?

I would say like, let's say for example, the infrared camera example, if you want to conduct similar sustainability, or not sustainability, but similar physical simulation for certain attributes.

@30:36 - **Tillman Sherwood**

Yeah, I think, I mean, I think this is also, I worked at, I did an internship at Autodesk, and I know that Autodesk is putting a lot of effort into adding this to their technology wherever possible.

So I think, I mean, I haven't used any of these tools in, since uni, to be honest. So in, what's.

Several years, but I think actually that the state-of-the-art technology, they were actually researching this when I was there, I think it's probably intermittent by now in some way, or it already was, but you can take the material, you can say how many grams of it you need, or you can even create simulations of the manufacturing process if you're using, for example, machining, if you're machining the part out of an aluminum, then it'll tell you how much waste, it creates.

And if you want to scale up, it'll tell you roughly, roughly the implications of your choice of manufacturing, but also of material and how bad it, like, like it gives it like a score, I think, or something based on different, different parameters.

So I think this is actually very, very easy with modern manufacturing tools. Also with SolidWorks, they must have it.

You have have it. And to answer your question specifically, I did do some simulations on the camera. I did an FEA simulation on the stress test, one of the aluminum parts.

But I didn't necessarily verify this with the actual part. I just was using it to get an idea of, if we make it this thin, you know, is it going to hold up?

And so I think this is a common thing for engineers to do, or nowadays, I think. Actually, maybe this is interesting for you as well.

I worked with an elderly, almost in retirement, automotive engineer. And I asked him, how does he determine, because he works with a very old version of Creo, a different CAD tool.

And, and he said, asked him, how did, how do you figure out the thickness of. . How thick that this should be or that should be and he's like, well, it's just experience.

I've been doing this for 30 plus years. And so if I think that this should be this thick, then it's just a gut feeling, I make it two millimeters thicker, and then we're done.

So it's interesting that there's this balance, I think, oftentimes, I think, maybe older generation engineers will just create the thing and make it good enough for the or depending on the tool, right, but it needs to be as good as it is.

But I think maybe younger engineers now are being taught in uni already, you need to try to make this as efficient as possible, and as thin as possible.

And I think what using these FVA tools, it can actually be pretty easy, I think, to quickly run a simulation, I've never verified these simulations, but I'm assuming they work well, because I'm sure they're used also in precision engineering industries like race cars,

Nice.

@34:06 - **Mouaz Al Qudsi (mouazq@gmail.com)**

That's a really cool explanation you gave about how old generation engineers work versus the current state. Nice, nice. So the tools you use right now, do you use Solidworks?

@34:28 - **Tillman Sherwood**

Yeah, I used Solidworks at this last company. I used Solidworks at Clyde HSI, the camera company. And I think I'm to be using Solidworks at my medicine company as well.

Oh, okay.

@34:41 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So you have the liberty or the freedom to choose the tools also you work with?

@34:50 - **Tillman Sherwood**

Actually, I think they will choose it. I think it's standardized at the company. I'm actually not sure what they use.

Amen. At the following company, but actually, also at the last one, we started with SolidWorks, and then we switched from SolidWorks to a tool called Onshape.

It's an online, kind of like Fusion 360. It's like on the web.

@35:18 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So if it might be interesting for you, I can also send you this add-on that SolidWorks developed. I think they developed it a really long time ago, and it makes products more sustainable.

But yeah, I don't know how useful, actually, those add-ons were.

@35:49 - **Tillman Sherwood**

Yeah, that's the thing. Sorry, I'm not able to... That's the thing. It always seems like it's... I've never actually heard of anyone using it, let's put it that way.

Yeah. So that's the thing. mean, the engineer that I spoke with that has the most experience in my life.

He just said like, yeah, it's a gut feeling because he has so much experience. I'm actually also curious to know if people use this in industry to make decisions.

But I mean, I'm sure it's relevant to just consider if you're a sustainability engineer at a big company, then, you know, it's good to question probably people's design choices with using these tools, right?

So, yeah, thank you for that.

@36:35 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I mean, it's just, yeah, I mean, I think they launch a lot of add-ons usually and what works works and what works doesn't work.

There's many stuff that, yeah, I think it's a big ecosystem, so no one is so good to be using.

I also work as a developer and there's always things I've never heard of, but companies launching as libraries and tools and things like this.

@36:59 - **Tillman Sherwood**

true. It's a very deep topic.

@37:01 - **Mouaz Al Qudsi (mouazq@gmail.com)**

I also found it interesting that Autodesk launched something called Generative Design. Yeah.

@37:08 - **Tillman Sherwood**

I feel I have tried it. Yeah, I don't think I've specifically tried it, but yeah, they use it regularly at Autodesk, I think.

It creates these very crazy-looking patterns that basically need to be additive manufactured, but there's certain places where it's already used.

There's a bridge in Amsterdam, I think, that's made by Autodesk from using Generative Design. Oh, I didn't know that.

But I think it's specifically used for these interesting applications. Yeah, I want to show this. means that. lot. Yes, back.

Ok, Thanks Here's the bridge. So, I mean, it's just a complete plug for Autodesk and Fusion 360, but, you know, it's completely additive manufactured.

Like you can see the lines. Yeah.

@38:18 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Wow. So, I mean, I don't know.

@38:22 - **Tillman Sherwood**

I feel like this is more and probably less sustainable than a normal bridge because it's 3D printed and welded together.

And also it's completely steel. So solid steel. mean, steel is also expensive from an environmental standpoint. And I mean, look at the thing must weigh so much.

I mean, guess it's hollow, but it's probably easier to make a wooden bridge with three steel beams or whatever.

So it's always interesting. mean, like.

@39:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Yeah, mean, in Autodesk original blog, when they developed this tool, which is interesting, it was even before LLM was a thing, they developed it, I think, some months ago before the OpenAI launched their ChatGBT commercial version, or, you know, the version, they launched it to the world.

And, I mean, Autodesk, they, in the blog was, this is, like, the generative design was supposed to be something for sustainability, and I think the idea was more probably that you see different alternatives of design, and then you see which one is more maybe sustainable, but, yeah, I mean, as far as I'm concerned, also, like, it didn't, it didn't get enough hype, so people didn't really use it.

time, there's know, know, So in which stage of product development you feel that you, or in which area you feel you can influence design decisions the most, if I, if you were, yeah.

@40:18 - **Tillman Sherwood**

Yeah, think in the, I'm trying to think, I guess, I guess it would be the concept phase. I mean, I think it's interesting with this last project that I worked on, there was this, I kind of showed you a picture already, this robot system.

We actually spent too much time on the design phase, on the, on the concept phase, I mean, because it was a very short project.

So it's just like a design that involved this robot in the center grabbing. Thank you. you. you. Moulds and putting them in the printer and then taking them out of the printer and putting them into the next conveyor belt and repeating that and so it was actually very fast paced project but what's interesting is that I had to like a debrief with my manager after the project and he said that yeah, he actually would have spent less time on the concept phase.

But I think in the concept phase it's important because I mean our concept phase is like two, three weeks maybe, but we were kind of discussing like what is it actually going to do and I think the concept phase is often crucial to create a create a requirements list that everyone agrees on.

I was just interviewing with a company that during the interviews they were all talking about different requirements and it was interesting to me, it's like well if I'm going to be making this thing then I

I don't want to, you know, be making only his requirements and not yours, the other guy. And, you know, I would have to create a list and we would have to create priorities so that we can, you know, meet the, um, make everyone happy, basically.

And I think it's the same with sustainability. I think if, if you don't consider sustainability from the beginning of the process, um, then, you know, it kind of is forgotten and it doesn't, it's not relevant.

Um, but at the same time, this shouldn't be used as a reason to with like hold back projects. I think it's important to get started and, and iterate if you can.

Um, so like, like you said, I think the design phase is also very important, but I think considering sustainability from the very beginning, um, keeping in mind that like this material is maybe less sustainable than other materials or solution is less sustainable than other solutions.

Um, when considering the requirements already we're creating a sustainable requirements.

@43:02 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Yes, and thank you so much for mentioning this. You're touching upon it because this is a question that I was about to exactly ask next, which is like, yeah, how do you track requirements on the product?

How do you share those requirements across departments or stakeholders? And your answer was like, you create a list and then you prioritize things.

And it's like, Tony, it's hard to, it's hard to agree on requirements. Yeah, yeah, I think so.

@43:27 - **Tillman Sherwood**

think that's one of my main learnings so far in industry is that the requirements are key and they can't be under addressed because it was interesting.

The last company, we tried to create a requirements list from my elderly engineer's experience. He wanted a signature so that the CEO could not change the requirements because in his experience, especially in startups, the requirements change.

Yeah. And it's important for the designer to. have a set list of requirements that don't change. So he was trying to get a signature, and I don't think the CEO wanted to sign anything.

And so we ended up, I created a list, and then they reviewed it actually twice. And then the product manager created a different list.

And then, you know, let's see, we had a, we had a, yeah, it was just, you know, we had a bunch of different lists, and we had some agreement.

And then at the end, we just ran with it, we had like a rough idea of what we all wanted to do.

And then we just, you know, we ran with this idea. And yeah, it's, it's, I think a little bit unofficial the way we did it, but it was just a Word document, really, with different priorities set.

And also, like, I think we attached numbers to the different priorities. And that's, that's a, this...

@45:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Also how it went during your previous experiences, where you worked on the infrared camera, for example? Yeah, with the camera, was similar.

@45:10 - **Tillman Sherwood**

We created a requirements list, but then he actually changed it during the project very often. And I think that is another reason why, for me, requirements has always been a very big thing in my brief career so far.

I think it's important to nail down the requirements in the beginning. Because otherwise, you're always creating new things and researching new things, and it's hard to not get stuck with the design phase.

@45:51 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Yeah. I mean, so we talked about sustainability, but... I know that it wasn't really a lot on the plate, but maybe I want to dive deep into a circular economy concept, like repairability, reusability, recyclability, stuff which you touched a little bit on the total energy use before.

But like for those other concepts, like the R strategies or things like, do you feel like you need, were you ever like in a place also where you thought, okay, I have to do an assessment on them or were also, they were relevant in a way?

@46:43 - **Tillman Sherwood**

So like, remind me again, reusability, what are the other Rs? Yeah, like repairability.

@46:54 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Repairability, yeah. Because, yeah, I mean, if you have a repairability requirement. and recyclability, so there must be a trade-off between them, so, for example, yeah.

@47:08 - **Tillman Sherwood**

Yeah, I think, unfortunately, there was too many things that we were doing at once to really dive deep into this topic.

I think it was always on our mind and trying to improve these things, but, yeah, I mean, it was kind of always on the forefront, but, I mean, for example, for repairability, they were designed to be, in the sense of the clay cups, they were designed to be broken, and they were designed to be cheap, so I think that played a role in it, that because it's cheap, you can throw it away, but I don't know if that actually helps sustainability in the end.

Something is designed to be cheap. true. And, I That's also an interesting thought. But yeah, in terms of a camera company, I did design it to be easily assembled and disassembled.

That's going be a factor. And so, yeah, it was designed to be easily taken apart and repaired and replaced.

That's only a consideration. And in terms of reusability for the clay company, was a situation because we had this dishwasher test that we were trying to do to put a number on how many times you can actually reuse it in a dishwasher.

And then recyclability. Yeah, I already touched on that. Yeah. So. So.

@49:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

You mentioned with the infrared camera, it was designed for disassembly, was that a requirement or you just did it?

@49:13 - **Tillman Sherwood**

Yeah, that was a requirement to make it easily assemblable because each item was hand-assembled by the engineers because these are quite expensive measurement tools.

It's like almost like 50,000 100,000 euros, I would estimate, would be the cost of one of these cameras. Yep.

So each machine would be hand-assembled and therefore it needs to be easily accessed, no small corners. And that's why also we designed this PCB to be able to plug everything in and call it a day, basically.

@49:56 - **Mouaz Al Qudsi (mouazq@gmail.com)**

And for that requirement, did you feel that the tools that you were using? Have you with assistance to this or did you feel that you needed experience with it?

Yeah, think it was more that I needed experience with it.

@50:11 - **Tillman Sherwood**

I often relied on their experience from assembling their previous machines. I think that was also a challenge just to imagine the assembly.

mean, this was the best tool that I could find. It was really just like taking it apart in CAD.

But often in CAD, you can drag things through each other. And so it's not really easy to imagine because you can create situations where you enclose something that's not physically possible to get there.

Does that make sense?

@50:52 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So, yeah, from what I understood from you, if I can repeat it, you tried to take the parts. of phrases who

And did you think about using any plug-ins, for example, or...?

@51:09 - **Tillman Sherwood**

I don't think so, no. I just, yeah. I wasn't aware of any plug-ins at the time, I think. Okay.

@51:18 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So kind of more improvising process. Yeah. And do you feel like there's a... you would have used, for example, if there would be a plug-in, or...?

Or, I mean, did you feel like it was painful to get to this requirement, or because of the experience of your colleagues, it was fine?

I definitely used...

@51:42 - **Tillman Sherwood**

or I would have used the plug-in, yeah, if it was available, because this was actually a constant factor in the design that re...

that it caused new problems because something didn't fit here or didn't fit there, and there was a... Yeah, if there was a way to, you know, simulate the assembly, that would be pretty cool, yeah.

@52:08 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Nice. I think we don't have, we have only ten minutes, and then this meeting would end on its own.

Is there, like, anything else regarding this topic you think you would love to tell me? Or if you have any questions?

@52:34 - **Tillman Sherwood**

I don't think so, yeah. I think we covered a lot of it. mean, my most, yeah, most of experience was from GaiaStar, where everything was very quick and startup-paced.

But yeah, I'm still relevant with the way that they were trying to make the processes more efficient. And, yeah.

But I guess... Questions for you, like, are you, so you're, with Carbon13, going to create a white paper with this, or with the information that you're gathering, and how will this white paper be used then?

@53:20 - **Mouaz Al Qudsi (mouazq@gmail.com)**

So, my partner, like, René, René Reich, he studied at KU Luven, and he spent, like, the last five years on the topic of designing for circular economy.

And we met at Carbon13, and we decided to dig into that topic, but more on, not on the academic level, but on how companies and designers really use it.

And, and what is the experience of, of organization from... From management also to designers that got this topic. So we're starting a pre-venture lab to research it, and we're collecting requirements seeing what current tools are offering, what are the challenges of the designers, and where circular economy sits.

And also, for example, you mentioned also the goal was also to optimize, reduce the cost. For example, when you do the CUPS, and this also contributes to sustainability.

We want to compile everything in one result paper by mid-February. Cool. Yes, and there's another colleague of us. His name is Olaf.

He's also like... Also like... Doing this research with us and yeah, I mean, we're still learning honestly, so we, we haven't like really cracked the, like how it, how it does work exactly because you have a lot of big organization and you have a lot of tools.

Yeah. What is really, what is really obvious is like design for sustainability and circular design, this comes usually post-production phase and not the design and conceptual phase, at least like the tools that provided currently right now, the LCA and the PLM tools, which you have mentioned, they're usually just used for reporting purposes.

Um, and, and, and we just want to see like how designers would deal with it at the early phase because all those decisions should be made at the design.

Yeah.

@56:02 - **Tillman Sherwood**

Okay. So it's kind of like, like we were the delving into it a little bit with the, um, yeah, using these, using these tools and does anyone actually use them or, uh, is it just to feel better about the design that you already created?

Um, or kind of like the, um, the, the material selectors and these, um, softwares. Yeah. Yeah. Yeah.

@56:26 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Um, yeah. Like the solid work sustainability. I mean, there's a lot of tools out there. Um, they haven't been used like widely, but also I think the problem is much more, a bit like much more complicated because you have the, you have the, for example, the recyclability and repairability requirement and which one you, you trade off or optimize on.

Yeah. Um, so, um, mean, you're in Berlin, right? So, um, yeah, you're also busy in Berlin and, uh, René is in Vienna.

But if you would like to know more, we can also schedule another meeting with them, maybe next year. We can also go for coffee or lunch, talk about it.

Yeah, I mean, we'll be happy, actually, to learn more. And yeah, we would love also to create a community for this topic here.

@57:31 - **Tillman Sherwood**

Yeah, I'm very interested in it. I'd love to learn more. But yeah, I'm traveling quite a bit during January, and then I start working in February.

But let's see, yeah.

@57:44 - **Mouaz Al Qudsi (mouazq@gmail.com)**

Nice, nice. Yeah, thank you so much for your time. I really appreciate it.

@57:49 - **Tillman Sherwood**

Yeah, no problem.

@57:50 - **Mouaz Al Qudsi (mouazq@gmail.com)**

You gave me the time for this. I'm very happy that we get to talk about this. Yeah.

@57:56 - **Tillman Sherwood**

Yeah. Yeah. this touch, think you posted.

@58:00 - **Mouaz Al Qudsi (mouazq@gmail.com)**

If we have any new findings and other stuff, if you know anybody, also would be interested in telling us more about this from your network, like someone that has been into this topic or also into design engineering and would love to, I would be happy to reach out to them.

Cool.

@58:25 - **Tillman Sherwood**

Yeah, I can maybe give one friend. I'll ask because he's in design engineering, but I'm not sure if it's relevant at his company.

I'll ask him and I can make the connection. Thank you so much. Have a great day. Yeah, thanks. You too.

Ciao. Ciao. Thank